LESSON 5.5

Determine the Volume of Rectangular Prisms

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism with whole-number edges using length × width × height.

WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Volume *Spanish: Volumen*	How much space is inside a solid shape.	V = l × w × h
Rectangular prism *Spanish: Prisma rectangular*	A solid box shape with six flat rectangle sides.	a cereal box
Cubic units *Spanish: Unidades cúbicas*	The units used to measure space inside, like cubic inches.	cm³, in³, ft³
Length, width, height *Spanish: Largo, ancho, altura*	How long, how wide, and how tall a box is.	V = l × w × h: for a box 5 × 3 × 2, volume = 30 cubic units
Net *Spanish: Plantilla (desarrollo plano)*	A flat shape that folds up into a solid.	A cross-shaped pattern of 6 rectangles folds into a rectangular box
Edge *Spanish: Arista*	The line where two flat sides of a solid meet.	A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A box is 6 cm long, 3 cm wide, and 4 cm tall.

1. One layer: 6 × 3 = 18 cubes.
2. 4 layers stack up.
3. 18 × 4 = 72.

Answer: V = 72 cm³

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller box: 5 long, 2 wide, 2 tall. What formula do we use?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The length is _____ , the width is _____ , and the height is _____ .
- The base area is _____ × _____ = _____ square units.
- The volume is _____ cubic units.
- I used cubic units instead of square units because _____ .

Word bank volume | rectangular prism | length | width | height | base area | layer | unit cube | cubic units | multiply | solid | face

WATCH OUT!

A common mistake

A common mistake in Volume with Whole Number Edges is multiplying only two of the three edges and stopping there

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the volume of each rectangular prism.

Length	Width	Height	Volume
5 in	3 in	4 in	
6 cm	6 cm	2 cm	
9 ft	2 ft	5 ft	

STEPS TO TRY

1. Each row gives a length, width, and height. The empty Volume cell wants one number with cubic units.
2. For each row use V = l × w × h: multiply the first two numbers, then multiply by the third.
3. Row 1: multiply 5 × 3 = 15 first, then multiply by the height, 4. Don't forget in³ in your answer.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the volume of a rectangular prism with l = 7 in, w = 3 in, h = 4 in?

(A) 14 in³
(B) 42 in³
(C) 84 in³
(D) 84 in²

STEPS TO TRY

1. Notice the three edge lengths: l = 7 in, w = 3 in, h = 4 in. Volume asks how much space fills the whole box.
2. Use the volume formula V = l × w × h, and remember volume is measured in cubic units (in³).
3. Start with 7 × 3 = 21 for the base, then multiply that result by the height, 4 inches.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A time capsule box has a volume of 120 cm³. Its length is 10 cm and width is 4 cm. What is its height?

(A) 3 cm
(B) 4 cm
(C) 6 cm
(D) 12 cm

STEPS TO TRY

1. You already know the volume (120 cm³) and two edges (10 cm and 4 cm). The height is the missing piece.
2. Work backward: since V = l × w × h, divide the volume by the base area (length × width) to find the height.
3. First find the base area: 10 × 4 = 40 cm². Then figure out what number times 40 gives 120.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Which unit is used for volume?

(A) Square inches (in²)

(B) Cubic inches (in³)

(C) Inches (in)

(D) Degrees (°)

STEPS TO TRY

1. Re-read each unit and think dimensions: volume fills 3D space — length, width, AND height.
2. Match the exponent to the dimensions: 2D area uses square units (²), so 3D volume needs one more dimension.
3. Picture stacking little cubes inside a box. Each cube is 1 in × 1 in × 1 in — which unit describes a cube like that?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Container C holds the most (96 in³) and Container B holds the least (50 in³). What do you notice about the dimensions that make one volume larger than another?

SENTENCE STARTER Container C has the largest volume because ___ × ___ × ___ = ___. Compared to Container B, the dimensions ___ produce a ___ volume because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help